A constructive theory of dilators

Abstract

We present a constructive version of part of Girard's theory of dilators (*Proof Theory and Logical Complexity*, chapters 8 and 9): the normal form, the extension to linear orders, the well-foundedness of the dilators built by sum, product and exponentiation in base 2, the link with the slow growing and Hardy hierarchies up to the Howard ordinal, morphisms, connected components and kinds. All objects are finite or decidable, well-foundedness is expressed by accessibility, and no proof uses the excluded middle. Everything has been checked in type theory, without axioms (function extensionality is used, as a hypothesis, only for two statements about trees of level Ω).

1 Introduction

Girard presents the theory of dilators in a classical setting: the category of ordinals, functors preserving direct limits and pull-backs, well-orders. Many of the arguments are nevertheless finite and combinatorial, and Girard's normal form (a point of $F(x)$ is written uniquely as $(z; x_0, \dots, x_{n-1}; x)_F$) suggests that the whole theory reduces to finite, decidable data. The aim of this note is to make this explicit.

The following choices are made throughout.

- We work on the category Δ_{inj} of finite ordinals with strictly increasing maps, coded by words. A pre-dilator is a functor $\Delta_{\mathrm{inj}} \to \Delta_{\mathrm{inj}}$ preserving pull-backs. It is thus given by finite data in each degree, and every property of it in a fixed degree is decidable.
- A linear order is a type with a decidable comparison with values in $\{<, =, >\}$. The extension of a pre-dilator to such an order is again a decidable linear order.
- Well-foundedness is accessibility: x is accessible if all its predecessors are, as an inductive definition. An order is well-founded if all its points are accessible. The absence of infinite descending sequences is never used.
- A disjunction is asserted only when we can tell which case holds. Where Girard concludes "$F' = 0$ or $F'' = 0$" with no possible decision, we state the negative form.

We point out along the way where the constructive version differs from the book.

2 Strictly increasing maps as words

Write $[n] = \{0, \dots, n-1\}$. A strictly increasing map $f : [m] \to [n]$ is determined by its image, hence by the word $w \in \{o, i\}^n$ whose letter y is i exactly when $y \in \mathrm{im} f$. Conversely every word w of length $|w| = n$ with $\#w = m$ letters i is a strictly increasing map $[m] \to [n]$. On words one defines by recursion:

- the composite $v \circ w$ (when $\#v = |w|$): replace the k-th letter i of v by the k-th letter of w;

- the intersection $v \wedge w$ of two words of the same length, letter by letter; it is the pull-back of v and w in Δ_{inj};
- the rank $\mathrm{rk}_w(y)$, the number of letters i before position y (the preimage of y), and the selection $\mathrm{sel}_w(z)$, the position of the z-th letter i (the map w itself);
- the inclusion $v \leq w$ of images.

Everything is decidable. The basic lemmas are: sel_w is strictly increasing on $[\#w]$, $\mathrm{sel}_{v\circ w} = \mathrm{sel}_v \circ \mathrm{sel}_w$, $\mathrm{sel}_w(\mathrm{rk}_w(y)) = y$ if $y \in \mathrm{im}w$, two words of the same length and the same image are equal, and a word w such that $s \circ w$ has the same image as s is the identity.

3 Pre-dilators and normal form

Definition 3.1. A *pre-dilator* is given by $D_0 : \mathbb{N} \to \mathbb{N}$ and a map D_1 from words to words such that $|D_1 w| = D_0|w|$, $\#D_1 w = D_0(\#w)$, $D_1(\mathrm{id}_n) = \mathrm{id}_{D_0 n}$, $D_1(v \circ w) = D_1 v \circ D_1 w$ and $D_1(v \wedge w) = D_1 v \wedge D_1 w$.

Remark 3.2. Girard's Definition 8.2.16 also requires monotonicity: $f \leq g$ implies $F(f) \leq F(g)$. We do not need it. For a dilator it follows from well-foundedness (8.2.15), by an argument on ω^{2y}; everything below holds without it.

Let $x \in D(p) = [D_0 p]$. A *support* of x is a word $u : [k] \to [p]$ with $x \in \mathrm{im}D_1 u$.

Theorem 3.3 (normal form, Girard 8.2.3)**.** *1. The supports of x are closed under intersection, and x has a least support* $\mathrm{supp}(x)$*, which is computable.*

2. The point $z = \mathrm{rk}_{D_1(\mathrm{supp}x)}(x)$ of $D(n)$, $n = \#\mathrm{supp}x$, is in the trace $\mathrm{Tr}(n)$*: it is in the image of no $D_1 v$ with $v : [k] \to [n]$, $v \neq \mathrm{id}$.*

3. If $x = \mathrm{sel}_{D_1 w}(z')$ with $z' \in \mathrm{Tr}$, then $w = \mathrm{supp}(x)$ and $z' = z$.

Proof. (1) If $x \in \mathrm{im}D_1 u \cap \mathrm{im}D_1 v$, then $x \in \mathrm{im}(D_1 u \wedge D_1 v) = \mathrm{im}D_1(u \wedge v)$: this is the only use of pull-backs. To compute the least support, let $g(u) = u$ if u is a support and $g(u) = \mathrm{id}_p$ otherwise (this is decidable), and

$$\mathrm{supp}(x) = \bigwedge_{u \in \{o,i\}^p} g(u).$$

The intersection ranges over a *finite* set of words; it is a support by (1), and it is contained in every support u since $g(u) = u$. (2) If $z = \mathrm{sel}_{D_1 v}(z'')$, then $\mathrm{supp}(x) \circ v$ is again a support of x, hence contains $\mathrm{supp}(x)$, and $v = \mathrm{id}$. (3) The word w is a support, so $\mathrm{supp}(x) = w \circ r$ for some word r; and $\mathrm{rk}_{D_1 w}(x) \in \mathrm{im}D_1 r$ forces $r = \mathrm{id}$ by the hypothesis on z'. □

Remark 3.4. In Girard, $\mathrm{supp}(x)$ is the intersection of all supports in ON, an infinite family. On Δ_{inj} it is indexed by the 2^p words of length p, and the normal form becomes an algorithm. In particular the trace is decidable: $z \in \mathrm{Tr}(n)$ if and only if $\mathrm{supp}(z) = \mathrm{id}_n$.

Corollary 3.5 (denotation form)**.** *The maps $x \mapsto (\mathrm{supp}x, z)$ and $(w, z) \mapsto \mathrm{sel}_{D_1 w}(z)$ are inverse bijections*

$$D(p) \;\cong\; \sum_{n \leq p} \mathrm{Tr}(n) \times \Delta_{\mathrm{inj}}([n], [p]).$$

4 Extension to decidable linear orders

Let X be a decidable linear order. A strictly increasing map $[n] \to X$ is a sorted list of length n.

Definition 4.1. $D(X)$ is the set of *denotations* (a, z), where a is a sorted list of X and $z \in \mathrm{Tr}(|a|)$. To compare (a, z) and (a', z'), form the merge c of a and a', with $a = c \circ u$ and $a' = c \circ u'$, and compare $\mathrm{sel}_{D_1 u}(z)$ and $\mathrm{sel}_{D_1 u'}(z')$ in $\mathbb{N}$.

Theorem 4.2. *1. The result does not depend on the common refinement c.*

2. *The comparison makes $D(X)$ a decidable linear order.*
3. *A strictly increasing $f : X \to Y$ induces $D(f)(a, z) = (f \circ a, z)$, strictly increasing; D is a functor, and the image of $D(f)$ consists of the denotations whose support lies in the image of f (preservation of pull-backs).*
4. *Every denotation is the image of a point of $D(n)$ along its own list (direct limits). For $X = \omega$, the denotations bounded by k form a copy of $D(k)$.*

Sketch. Two common refinements factor through their merge, and a sorted list is injective; one concludes by $\mathrm{sel}_{D_1(ev)} = \mathrm{sel}_{D_1 e} \circ \mathrm{sel}_{D_1 v}$ and the strict monotonicity of $\mathrm{sel}_{D_1 e}$. Transitivity is read in a common refinement of the three lists. If two denotations give the same point of $D(|c|)$, they are equal by the denotation form. □

5 Well-foundedness

Definition 5.1. For a decidable relation $<$ on A, accessibility Acc is the least predicate such that x is accessible as soon as all $y < x$ are. A is *well-founded* if every point is accessible. A pre-dilator D is a *dilator* if $D(X)$ is well-founded for every well-founded decidable linear order X.

This definition quantifies over all orders X in a universe; it is a statement of a higher universe. The induction principle associated with Acc is the only form of transfinite induction used.

Lemma 5.2 (inverse image)**.** *If $\varphi : A \to B$ preserves $<$ and B is well-founded, so is A. The ordered sum and the lexicographic product of two well-founded orders are well-founded, and so is $\mathbb{N}$.*

Theorem 5.3. Id*, 1, $D + E$, $D \cdot E$ are dilators if D and E are.*

Proof. In each case $D(X)$ is mapped into a known order, preserving the order. The trace of Id is $\{0\} = \mathrm{Tr}(1)$, so $\mathrm{Id}(X) \cong X$. For the sum, $(D + E)(n) = D(n) + E(n)$ and the trace of $D + E$ is that of D followed by that of E, shifted; $(D + E)(X)$ maps into the ordered sum $D(X) + E(X)$. For the product, $(D \cdot E)(n)$ is $E(n)$ copies of $D(n)$, ordered by the Kronecker product of words. A point $z = e \cdot D_0(n) + j$ (Euclidean division) gives the points e of $E(n)$ and j of $D(n)$, which are pushed into $E(X)$ and $D(X)$; the order is lexicographic, E first. □

6 Exponentiation

6.1 2^X is well-founded

For a linear order Y, 2^Y is the set of finite subsets, compared by the largest element of the symmetric difference. It is represented by the strictly decreasing lists, compared lexicographically (the first element is the largest, and a proper prefix is smaller).

Theorem 6.1. *If Y is well-founded, so is 2^Y.*

Proof. The empty list is the least element, hence accessible. Let $P(x)$ be: every decreasing list with head x is accessible. We prove $P(x)$ by induction on the accessibility of x. Assume $P(y)$ for all $y < x$.

- Every decreasing list whose head is $< x$ is accessible: it is the empty list, or a list with head $y < x$.
- Let l be a decreasing list with elements $< x$; it is accessible by the previous point. We show, by induction on the accessibility of l, that $x :: l$ is accessible. A predecessor of $x :: l$ is the empty list, or $y :: k$ with $y < x$ (accessible by $P(y)$), or $x :: k$ with $k < l$ (accessible by the induction hypothesis on l).

Since every decreasing list is empty or of the form $x :: l$, all of them are accessible. □

Remark 6.2. Girard recalls (§9.6) that the statement "X well-ordered $\Rightarrow 2^X$ well-ordered" is equivalent, over a weak theory, to $(\Sigma^0_1\text{-CA})^*$. With accessibility as an inductive definition, the proof is an elementary double induction: the logical strength lies in the existence of inductive definitions, not in a comprehension principle.

6.2 The pre-dilator 2^D

On finite ordinals, 2^n is the number 2^n, a subset s of $[n]$ being the number $\sum_{k\in s} 2^k$. The order of numbers is then the order of $2^{[n]}$. For $w : [m] \to [n]$, $2^w : [2^m] \to [2^n]$ sends the subset s to its image $w(s)$; on words, position j is in the image of 2^w if and only if the subset j is contained in im$w$. The first letter of w corresponds to the least significant bit, which gives the recursion

$$2^{\varnothing} = i, \qquad 2^{o\,w} = \mathrm{lo}(2^w), \qquad 2^{i\,w} = \mathrm{dbl}(2^w),$$

where lo puts an o after each letter and dbl doubles each letter. Composition and intersection commute with lo and dbl, so 2^- is a pre-dilator, and so is $2^D = 2^- \circ D$.

Theorem 6.3. *If D is a dilator, so is 2^D.*

Proof. A denotation (a, z) of $(2^D)(X)$ is a subset s of $D(|a|)$ (the bits of z). We associate with it the decreasing list $\varphi(a, z)$ of the points $D(a)(t)$ of $D(X)$, for $t \in s$. By Theorem 6.1 and the inverse image lemma, it suffices to show that φ preserves the order. Let c be the merge of a and a', with $a = c \circ u$. Three combinatorial facts suffice:

1. the point of $2^{D(|c|)}$ image of s is the subset $D_1u(s)$, that is $\mathrm{sel}_{2^W}(\mathrm{num}\, s) = \mathrm{num}(W \circ s)$ for a word W;
2. comparing two numerals of the same length amounts to comparing lexicographically the decreasing lists of their bits equal to 1;
3. the bits equal to 1 of $W \circ s$ are the images by sel_W of those of s, in the same order.

Finally, two images $\mathrm{sel}_{D_1u}(t)$ and $\mathrm{sel}_{D_1u'}(t')$ compare as the points $D(a)(t)$ and $D(a')(t')$ of $D(X)$. □

Corollary 6.4. *The pre-dilators built from* Id *and* 1 *by* $+$, $\cdot$ *and* 2^- *are dilators; in particular $D(\omega)$ is well-founded.*

7 Trees and hierarchies

We use the Brouwer trees $\mathcal{O}$ (0, successor, limit of a sequence) and $\mathcal{O}_2$ (in addition, limit of a family indexed by $\mathcal{O}$), with the slow growing hierarchy $G_1 : \mathcal{O} \to \mathbb{N} \to \mathbb{N}$, the Hardy hierarchy H_1, and their analogues $G_2 : \mathcal{O}_2 \to \mathbb{N} \to \mathcal{O}$, $H_2 : \mathcal{O}_2 \to \mathcal{O} \to \mathcal{O}$.

With each polynomial dilator p we associate a tree $t(p)$: $\mathrm{Id} \mapsto \omega$, $1 \mapsto 1$, sum and product of trees, and 2^b with $2^{b+1} = 2^b + 2^b$. We have

$$G_1(t(p))(k) = |D_p(k)|,$$

which is Girard's statement 9.A for these dilators: the slow growing hierarchy along $D(\omega)$ gives back D. The tree $t_2(p) \in \mathcal{O}_2$ (where $\mathrm{Id} \mapsto \Omega$) satisfies the condition on fundamental sequences needed for Wainer's collapsing theorem, and $G_2(t_2(p))(n) = t(p)$.

Theorem 7.1. *Let $T_0 = \mathrm{Id} + 1$ and $T_{k+1} = 2^{T_k}$. The limits of the trees are $\varepsilon_0 = \sup_k t(T_k)$ and $\varepsilon_{\Omega+1} = \sup_k t_2(T_k)$; let $\eta_0 = H_2(\varepsilon_{\Omega+1})(\omega)$ (the Howard ordinal, in Girard's sense, 9.A). Then, for all n,*

$$G_1(\eta_0)(n) = H_1(\varepsilon_0)(n).$$

Function extensionality is used in one place only: the commutation of G_2 with the arithmetic under the limits indexed by $\mathcal{O}$. It is taken as a hypothesis.

8 Morphisms

Definition 8.1. A natural transformation $T : D \to E$ is a family of words $t_n : [D_0 n] \to [E_0 n]$ such that $E_1 w \circ t_m = t_n \circ D_1 w$ for $w : [m] \to [n]$.

Proposition 8.2 (Girard 8.2.5 (ii))**.** *If $z \in \mathrm{Tr}_D(m)$, then $T(m)(z) \in \mathrm{Tr}_E(m)$, and T sends the denotation (w, z) to $(w, T(m)(z))$.*

Proof. Let $v : [k] \to [m]$ with $T(m)(z) \in \mathrm{im} E_1 v$. The equality $v = \mathrm{id}$ is decidable; assume $v \neq \mathrm{id}$. Then v misses a point p, and the two injections $g_1, g_2 : [m] \to [m+1]$ that skip p and $p+1$ are distinct and agree on v. So $E(g_1)$ and $E(g_2)$ agree on $T(m)(z)$, and by naturality and injectivity of $T(m+1)$, $D(g_1)(z) = D(g_2)(z)$. Since z is in the trace, g_1 and g_2 are both the least support of this point (Theorem 3.3), a contradiction. Hence $\neg\neg(v = \mathrm{id})$, and $v = \mathrm{id}$ by decidability. □

Proposition 8.3 (Girard 8.3.5 (ii))**.** *Two natural transformations $D \to E$ that agree on the traces are equal.*

Proof. Every $x \in D(n)$ is $\mathrm{sel}_{D_1 w}(z)$ with z in the trace, and $T(n)(x) = \mathrm{sel}_{E_1 w}(T(m)(z))$. Two words of the same length with the same values under sel are equal. □

The inclusions of D and of E into $D + E$ are natural; their traces are $z \mapsto z$ and $y \mapsto D_0(m) + y$.

9 Connected components

Definition 9.1. Two trace points (m, z) and (m', z') of D are *separated* if, for all words u, u' of the same length with $\#u = m$ and $\#u' = m'$, we have $\mathrm{sel}_{D_1 u}(z) < \mathrm{sel}_{D_1 u'}(z')$. They are *linked* if they are separated in neither order.

Being separated means that every denotation with trace (m, z) is below every denotation with trace (m', z'), in $D(X)$ for every X, whatever the coefficients: this is $\S_D(m, z; m', z') = (0, +1)$ in Girard (8.4.18).

Definition 9.2. On Δ_{inj} an increasing bijection of $[n]$ is the identity, so two isomorphic pre-dilators are equal. We write $D = F + G$ if $D_0 n = F_0 n + G_0 n$ and $D_1 w = F_1 w \cdot G_1 w$ (concatenation of words) for all n and w. D is *non-zero* if there are n and a point of $D(n)$. D is *connected* if it is non-zero and, for every decomposition $D = F + G$, F and G are not both non-zero.

Remark 9.3. Girard requires "$F = 0$ or $G = 0$". The statement $F = 0$, that is $\forall n\ F_0 n = 0$, is not decidable, and a pre-dilator that is zero in degree n may be non-zero in a higher degree. The negative form is the right constructive version; classically it is equivalent.

Theorem 9.4 (Girard 9.1.7)**.** *If F and G are non-zero, $F + G$ has two separated trace points. Consequently a non-zero pre-dilator whose trace points are pairwise linked is connected.*

Proof. A point of $F(n)$ has a normal form with trace (m, z), a point of $G(n')$ a normal form with trace (m', y). In $F + G$, (m, z) and $(m', F_0 m' + y)$ are trace points (images under the inclusions), and for u, u' of the same length L,

$$\mathrm{sel}_{(F+G)_1 u}(z) = \mathrm{sel}_{F_1 u}(z) < F_0 L \leq F_0 L + \mathrm{sel}_{G_1 u'}(y) = \mathrm{sel}_{(F+G)_1 u'}(F_0 m' + y). \qquad \square$$

Example 9.5. Id and 1 each have a single trace point, $(1, 0)$ and $(0, 0)$; they are connected. $\mathrm{Id} + 1$ is not.

The construction of the components rests on the restriction to a set of trace points.

Theorem 9.6 (Girard 8.3.6)**.** *Let c be a decidable set of trace points of D. Let $F(n) = \{x \in D(n) \mid$ the trace of x is in $c\}$ and $F(w)$ the restriction of $D(w)$. Then F is a pre-dilator, the inclusion $F \to D$ is a natural transformation, and its trace is contained in c.*

Proof. Let $\chi_n : [F_0 n] \to [D_0 n]$ be the inclusion. The key point is that $D(w)$ preserves traces: if $x = D(s)(z)$ with z in the trace, then $D(w)(x) = D(w \circ s)(z)$, and by uniqueness of the normal form the trace of $D(w)(x)$ is $(\#s, z)$. It follows that F is a pull-back of D:

$$D_1 w \circ \chi_m = D_1 w \wedge \chi_n.$$

We define $F_1 w$ by $\chi_n \circ F_1 w = D_1 w \circ \chi_m$. The functorial identities are checked after composing on the left with χ_n, which can be cancelled, using for intersections that composition distributes over $\wedge$. $\square$

Theorem 9.7 (cut)**.** *Let c be as above, such that every trace point in c is separated from (and below) every trace point outside c. Then $D = D|_c + D|_{\neg c}$. In particular a connected pre-dilator has no proper cut.*

Proof. If $x, y \in D(n)$ have their traces in c and outside c respectively, then $x < y$: apply separation to the supports of x and y. The points of $D|_c(n)$ thus form an initial segment of $D(n)$, so that $\chi_n = \mathrm{id}_a \cdot 0^b$ and $\chi'_n = 0^a \cdot \mathrm{id}_b$. The pull-back property then identifies the two blocks of $D_1 w$ with $F_1 w$ and $G_1 w$. $\square$

Girard's Theorem 9.1.6 (unique decomposition of a dilator as a sum of connected components indexed by an ordinal) is obtained by cutting along the classes of the relation "linked"; Theorems 9.6 and 9.7 are its constructive building blocks. What remains is the transitivity of this relation (the ultrametric inequality 8.4.21), the well-ordering of the quotient, and the indexed sum.

10 Kinds

Girard classifies dilators into four kinds (9.1.10): 0; 1 ($F = F' + 1$); ω (a sum indexed by a limit ordinal); Ω ($F = F' + F''$ with F'' connected and different from 1). For a tree in $\mathcal{O}_2$, the kind is the head constructor: 0, successor, limit over $\mathbb{N}$, limit over $\mathcal{O}$.

Theorem 10.1. *Let p be a polynomial dilator (with 2^-). The tree $t_2(p)$ is of kind 1 or Ω, never 0 nor ω. It is of kind 1 if and only if, for some (equivalently every) n, the largest point of $D_p(n+1)$ is a constant, that is, a point coming from $D_p(0)$.*

For every F, the largest point of $(F+1)(n)$ is a constant; the theorem thus identifies, for polynomial dilators, kind 1 of the tree with Girard's kind 1. The kind of 2^D is 1 exactly when D is constant.

11 Finite and ideal elements

In constructive domain theory one works with the finite (compact) elements, and expresses results in terms of ideal elements, which are determined by their finite approximations. The theory of dilators has exactly this structure, and historically the two are related: Girard's stable domains and coherence spaces were obtained by simplifying dilators.

domain theory	*dilators*
compact elements	finite ordinals, Δ_{inj}, words
information system	the data (D_0, D_1) on words
ideal elements	$D(X)$, X any decidable linear order
continuity	preservation of direct limits
stability (Berry)	preservation of pull-backs
minimal approximation	least support
trace of a stable map	trace $\{(n,z) \mid z \in \mathrm{Tr}(n)\}$
points of the completion	denotations $(z; x_0 < \cdots < x_{n-1}; X)$

Every linear order is the directed union of its finite suborders, and a pre-dilator, which preserves direct limits, is determined by its restriction to Δ_{inj}. This is the analogue of a continuous map being determined by its action on compact elements.

Stability gives canonical names. In a Scott domain an ideal element is a class of approximations, with no canonical representative. Here, by Theorem 3.3, every point of $D(X)$ has a *least* finite approximation, its support. So an ideal point has a canonical name, the pair (support, trace point), and $D(X)$ can be defined directly by denotations, with decidable order and equality, instead of as a quotient. Preservation of pull-backs plays the role of Berry's stability condition, and it is what makes the ideal level as computable as the finite one. Conceptually $D(X)$ is the colimit of the $D(n)$ over the finite subsets of X (a left Kan extension along $\Delta_{\mathrm{inj}} \to$ linear orders); the denotations compute this colimit.

Proofs on finite elements, statements on ideal elements. All the proofs above are about finite objects: words, $D(n)$, the merge of two supports. Many of the statements are about ideal ones: $D(X)$ is a decidable linear order, functorial in X, well-founded when X is, and the hierarchy theorems are about $D(\omega)$ and $D(\Omega)$. The results on morphisms say that a natural transformation is determined by its action on traces (Proposition 8.3.5 (ii)), as an approximable mapping is determined on the basis. Even Ω is handled as an ideal ordinal: the tree of $D(\Omega)$ has limits indexed by all countable trees, and the collapsing theorem only uses it through its approximations.

Uniqueness of the ideal object. Call an *extension* of D to X an ordered type G with, for each finite sorted subset c of X, a map $\iota_c : D(|c|) \to G$ such that $\iota_c \circ D(w) = \iota_{c \circ w}$, ι_c is strictly increasing, and every point of G is some $\iota_c(x)$. (Preservation of pull-backs is not needed: it is already in D.)

Theorem 11.1 (Girard 8.2.6, univalent form)**.** *Assume function extensionality and univalence. Every extension G of D to X is equal, as an ordered type, to $D(X)$.*

Proof. The map $\varphi(a, z) = \iota_a(z)$ preserves the comparison: one computes in the merge c of two supports, using naturality and the monotonicity of ι_c. It is injective because $D(X)$ is linear, and surjective because $\iota_c(x) = \varphi(c \circ w, z)$, where (w, z) is the normal form of x in $D(|c|)$. An equivalence preserving the comparison is an equality of ordered types, by univalence and the structure identity principle. □

Univalence and function extensionality are hypotheses (types), not axioms, and nothing at the finite level depends on them. So the three levels are separated: the finite level is decidable, the ideal level is described by its finite approximations, and the univalent level turns "determined by" into equality.

Where Girard's text looks non-constructive. These are the places where an ideal element is treated as a completed whole: deciding the kind of a dilator, deciding $F = 0$, the cofinality of $F(\mathrm{On})$. The remedy is the one of domain theory: replace the global decision by a statement about approximations. Connectedness, stated negatively, is an example.

12 Summary

The above shows that a substantial part of Girard's theory is constructive, and often more simply so than in the book:

- the normal form is an algorithm, because the intersection of supports ranges over a finite family;
- the extension to linear orders and all comparisons are decidable;
- well-foundedness, expressed by accessibility, propagates by elementary inductions, including for 2^X;
- Girard's proofs by contradiction (8.2.5) concern decidable properties and can be eliminated;
- connectedness must be stated in negative form, and it is the only notion of chapter 9 treated here that is not decidable.

Not treated: 9.1.6 in full, kinds for an arbitrary dilator, the order type of $t(p)$ as the ordinal of $D_p(\omega)$, and beyond, Π^1_2- completeness, the well-ordering of dilators and the functor Λ.

A Correspondence with the formalization

The file `dilator.seg` (about 1350 declarations) is checked by a type-theory checker in which the only recursion is the elimination of inductive types, with no axioms.

Theorem 3.3 and corollary	`PreDil.Point`, `fwd`, `bwd`
Extension to X	`LinOrd.DenOrd`, `Fun`
Theorem 5.3	`IdWF`, `OneWF`, `SumWF`, `ProdWF`, `polDil`
Theorem 6.1	`Exp2.wfE`
Theorem 6.3	`ExpDil`, `ExpWF`
Trees, η_0	`hier`, `FE.hierH`, `FE.hierEta`
Morphisms	`Nat.trT`, `Nat.denT`, `NatEq.Pts.eqT`
Theorem 9.4	`ConnSum`, `ConnCrit.connected`
Theorem 9.6	`Restr`
Theorem 9.7	`SplitCut.isSum`, `noCut`
Kinds	`treeK`, `kindPol`, `kind1`
Theorem 11.1	`UA.sip`, `UA.Ext.Uniq.uniq`